

Harsh Desai

(224)-830-5306 · desaiharsh2006@gmail.com · [linkedin.com/in/harshdesai8](https://www.linkedin.com/in/harshdesai8) · github.com/HuskyONG

EDUCATION

University of Illinois Chicago

Expected May 2027

B.S. Computer Science

TECHNICAL SKILLS

Languages: Python, Java, C++, C, JavaScript, TypeScript, HTML/CSS, Swift, SQL, Lua, C#, Qiskit, Ruby, Next.js, Flask, JIRA, Django, Go

Frameworks & Tools: Claude Code, React, React Native (Expo), Next.js, Spring Boot, MongoDB, Firebase (Auth, Firestore), Git/GitHub, Jupyter Notebook, Qbraid, Xcode, Unity, IntelliJ, VS Code, AWS, Kafka, Kubernetes, Docker, Express.js, REST APIs, Android Studio, Agile

EXPERIENCE

Quantum Algorithms Researcher | University of Illinois Chicago

Aug 2025 – Mar 2026

- Investigated **Jordan-Wigner** vs. **Bravyi-Kitaev** fermion-to-qubit mappings to reduce circuit complexity by **75%**, benchmarking simulation efficiency across multi-qubit systems while maintaining equivalent time-control accuracy.
- Designed experimental frameworks in **Qiskit** to benchmark qubit mapping approaches, producing reproducible comparisons across multi-qubit quantum circuit configurations.

Quantum Software Development Intern | University of Illinois Chicago

Summer 2025

- Doubled measurement accuracy (**60%**) across variable configurations by building **Python** scripts to compute expectation values for Quantum Harmonic Oscillators and multi-qubit systems.
- Produced reproducible experimental results analyzing behavioral shifts in **Quantum Harmonic Oscillators** under both incremental and drastic variable changes.

Computer Science Instructor | Code Ninjas, Long Grove

2022 – 2024

- Delivered a full-stack development lifecycle curriculum for **100+** students aged 8–13, covering programming fundamentals, game design, and AI concepts using **Roblox Studio**, **Scratch**, and **Code Combat**.
- Created and iterated custom lesson plans and slide decks for rotating class cohorts, adapting material for varying skill levels based on student feedback.

PROJECTS

Splitz | React Native (Expo), TypeScript, Firebase (Auth, Firestore)

August 2026

- Built a cross-platform bill-splitting app on **React Native (Expo)** with a **TypeScript** codebase and **Firebase (Auth, Firestore)** backend, syncing group expenses in real time across **Android, iOS, and web** from a single codebase.
- Engineered custom and even-split expense logic with a **debt-simplification algorithm** to minimize settling transactions per group, plus push-notification-driven payment reminders via the **Expo push API**.
- Implemented **native Google Sign-In** via an **EAS custom dev client** alongside an **expo-auth-session** web fallback, and secured user data with locked-down **Firestore security rules** deployed via the Firebase CLI.

CurtainCall | Next.js, React, Spring Boot, MongoDB, Tailwind CSS

[Live Demo](#) Feb 2026

- Engineered a **Tinder-style** artist networking platform with a swipe-based matching system, shipping **12+ RESTful API endpoints** in **Spring Boot** to support user connections, endorsements, and discovery.
- Architected **MongoDB** collections around user relationships and endorsement logic, structuring schemas to support **1,000+ concurrent users** while eliminating redundant database calls.
- Built a responsive **Next.js/React** frontend styled with **Tailwind CSS**, implementing swipe gesture handling and optimistic UI updates for a smooth, mobile-first matching experience.

SnapLink | React, Spring Boot, MongoDB, Google Maps API

[Live Demo](#) Feb 2025

- Developed a geo-tagged task discovery platform connecting small businesses with local job-seekers in real time, integrating the **Google Maps API** with dynamic geolocation filtering for hyper-local opportunity matching.
- Implemented a full-stack architecture with **RESTful Spring Boot APIs** and **MongoDB**, enabling scalable task posting, status tracking, and location-based querying in **React**.
- Designed **geospatial MongoDB queries** to power radius-based search, letting job-seekers filter opportunities by distance without a linear scan of every posting.

CountingCards | HTML, CSS, JavaScript, Bootstrap

[Public Link](#) Feb 2026

- Created a gamified card-based study system with dynamic card cycling, interactive session tracking, and a responsive **Bootstrap** UI, replacing passive reading with active recall loops in **HTML, CSS, and JavaScript**.
- Built session persistence with **localStorage** so study progress and card decks survive page reloads, and added spaced-repetition-style cycling to resurface missed cards more frequently.

AWARDS & LEADERSHIP

- IBM Quantum Badge of Excellence — completed all labs in the **Qiskit Global Summer School 2025** with **100%** accuracy.
- Freshman Mentor Program — selected as 1 of **100** mentors from **2,000+** applicants; recognized as a **top mentor** by program leadership.
- IEEE Executive Board — **Web development board member**; leading redesign and modernization of the IEEE UIC chapter website.